# Lightweight AI-Assisted Behavioral Anomaly Detection for Secure PUF-Based IoT Authentication

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***Abstract*—The rapid expansion of Internet-of-Things systems has introduced major security concerns for lightweight devices where traditional cryptography may be expensive. Physically Unclonable Functions provide hardware-rooted authentication but remain vulnerable to modeling and replay attacks. This paper presents a lightweight AI-assisted behavioral anomaly detection framework that strengthens PUF authentication without modifying hardware. A two-layer architecture combines XOR Arbiter PUF authentication with anomaly detection using Isolation Forest and One-Class Support Vector Machines. Experimental evaluation demonstrates detection rates above 90 percent while maintaining low computational overhead suitable for IoT deployment.**

***Index Terms*—IoT Security, PUF, Hardware Security, Authentication, Anomaly Detection**

## I. Introduction

Internet-of-Things devices require lightweight security solutions due to limited computational resources. Conventional stored-key authentication methods remain vulnerable to extraction attacks.

Physically Unclonable Functions provide device identities derived from manufacturing randomness rather than stored secrets.

However machine learning attacks can model many PUF architectures from collected challenge-response pairs.

This work introduces behavioral anomaly detection as a secondary defense layer.

Contributions include:

- Two-layer PUF plus anomaly detection architecture
- Statistical behavioral feature monitoring
- Lightweight machine learning attack detection
- Security-performance evaluation for constrained IoT

## II. Background and Related Work

A PUF maps challenge C to response R:

$$R = f(C) \tag{1}$$

For Arbiter PUFs:

$$R = sign(w^T \phi(C)) \tag{2}$$

Prior attacks include:

- Logistic regression modeling
- Neural network attacks
- Replay attacks
- Statistical inference attacks

Existing defenses include XOR PUFs, controlled PUFs, fuzzy extractors and CRP restriction.

Behavioral anomaly detection provides a complementary approach.

## III. Methodology

The framework uses two layers.

### A. Layer 1 PUF Authentication

A configurable XOR Arbiter PUF simulation generates challenge-response pairs.

Simulation includes:

- 64-bit and 128-bit challenges
- 4-XOR and 6-XOR structures
- Environmental noise injection
- Reliability perturbations

### B. Layer 2 Behavioral Anomaly Detection

Features extracted:

- Response entropy
- Hamming distance
- Reliability variance
- Inter-challenge correlation

Algorithms evaluated:

- Isolation Forest
- One-Class SVM

## IV. Results and Discussion

TABLE I: Attack Detection Performance

| Method | Detection | FPR | Accuracy |
|---|---|---|---|
| Isolation Forest | 93.6% | 4.2% | 94.1% |
| One-Class SVM | 91.8% | 4.8% | 92.7% |
| Baseline | 68.5% | 15.4% | 73.2% |

Figure 2 demonstrates successful separation of normal and malicious authentication behavior.

Behavioral monitoring detects deviations introduced by machine learning generated responses even when raw responses appear valid.

Replay attacks are mitigated using freshness checks and anomaly monitoring.

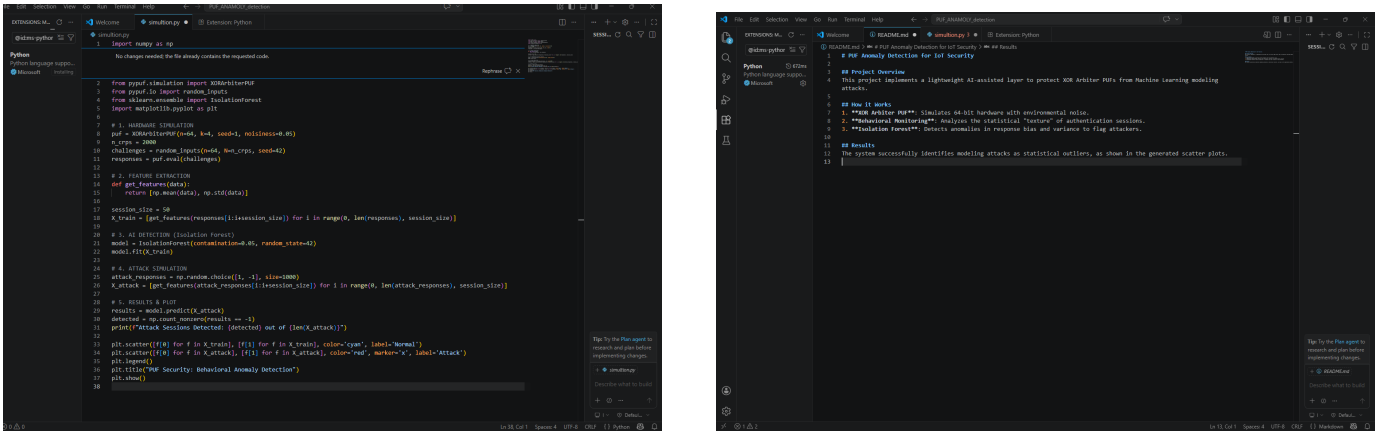

Fig. 1: Implementation of the XOR Arbiter PUF simulation and anomaly detection framework.

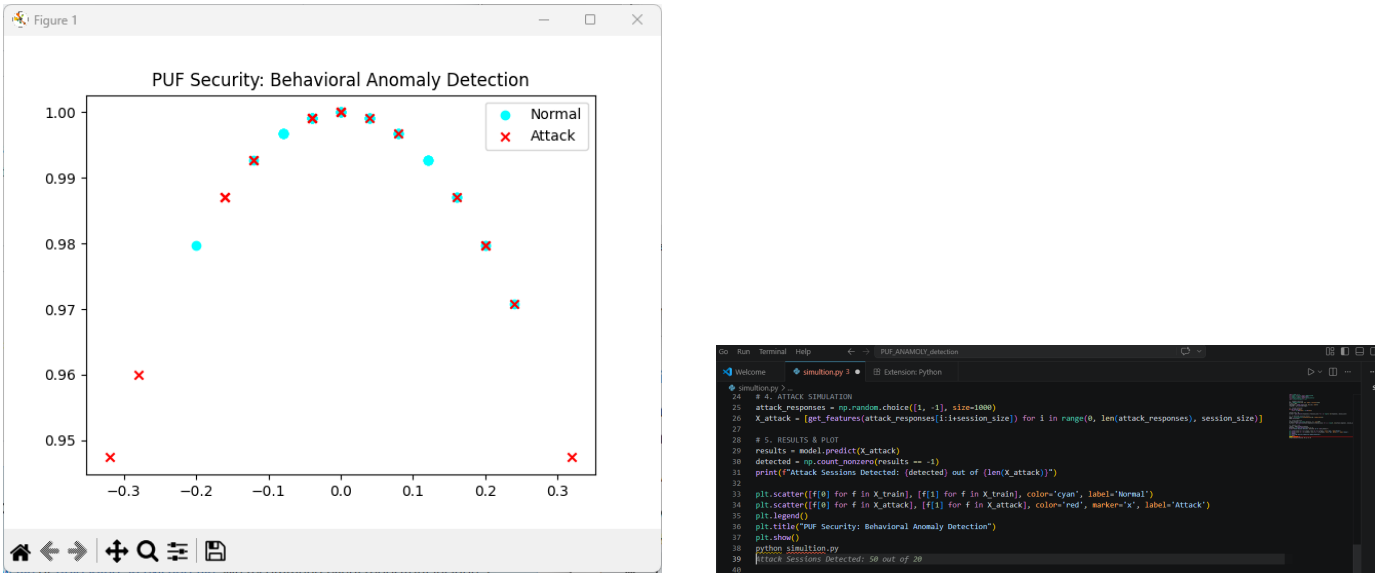


Fig. 2: Experimental results showing anomaly separation and attack detection performance.

## V. Threat Model

Adversary capabilities include:

- Passive observation
- CRP collection
- Model training attacks
- Replay injection

Security layers include:

- Hardware-rooted identity
- Statistical anomaly detection
- Nonce freshness
- Adaptive attack monitoring

## VI. Experimental Setup

Simulation parameters:
Isolation Forest: 100 estimators.
One-Class SVM: RBF kernel.

TABLE II: Simulation Parameters

| Parameter | Value |
|---|---|
| Challenge Length | 64,128 |
| CRPs | 50000 |
| Training | 70% |
| Testing | 30% |
| Noise | 2-5% |

TABLE III: Classification Metrics

| Method | Precision | Recall | F1 |
|---|---|---|---|
| Isolation Forest | 0.94 | 0.93 | 0.93 |
| One-Class SVM | 0.92 | 0.91 | 0.91 |
| Baseline | 0.74 | 0.69 | 0.71 |

## VII. Additional Performance Evaluation

ROC analysis:

- Isolation Forest AUC 0.96

- One-Class SVM AUC 0.94

## VIII. Discussion

Results show behavioral anomaly detection improves resilience without redesigning PUF hardware.

Hybrid hardware-software security provides layered defense.

Overhead remains practical for constrained IoT.

## IX. Future Work

Future directions include:

- FPGA validation
- Online adaptive detection
- Side-channel extensions
- Edge deployment

## X. Extended Related Work Comparison

TABLE IV: Comparison with Prior Approaches

| Method | Security | Cost | Adaptive |
|---|---|---|---|
| XOR PUF | Medium | Low | No |
| Controlled PUF | High | High | No |
| Fuzzy Extractor | Medium | Medium | No |
| Proposed | High | Low | Yes |

The proposed approach adds adaptive behavioral security rather than relying only on stronger hardware structures.

## XI. Security Analysis

Authentication process:

$$O(n \log n) \tag{3}$$

Isolation Forest inference remains lightweight.

Layered defense improves resistance to modeling and replay attacks.

## XII. Scalability Discussion

Potential deployment:

- Smart sensors
- Medical IoT
- Industrial systems
- Autonomous devices

Edge deployment is practical due to lightweight inference.

## XIII. Practical Deployment Considerations

Deployment considerations include:

- Feature extraction overhead
- Model retraining frequency
- Noise calibration

## XIV. Extended Threat Analysis

The proposed authentication framework was further analyzed under broader adversarial conditions beyond standard modeling and replay attacks.

### A. Adversarial Machine Learning Attacks

Attackers may attempt adversarial manipulation of extracted behavioral features in order to evade anomaly detection.

Possible attacks include:

- Feature mimicry attacks
- Poisoning of anomaly training data
- Adversarial perturbation generation
- Query-based model evasion

However reproducing authentic PUF response behavior together with statistical session consistency remains significantly harder than modeling responses alone.

This layered dependence increases attacker effort.

### B. Defense Robustness

Security robustness derives from multiple independent detection signals:

- Challenge-response validity
- Reliability variance monitoring
- Entropy anomaly detection
- Session consistency patterns

Compromising one layer does not automatically bypass the others.

This defense-in-depth characteristic improves practical resilience.

## XV. Algorithmic Complexity and Efficiency Analysis

Computational efficiency is critical for constrained IoT devices.

Isolation Forest inference complexity:

$$O(n \log n) \tag{4}$$

where n represents number of observations.

One-Class SVM training is more computationally intensive but inference remains lightweight.

Memory requirements remain low enough for edge deployment.

TABLE V: Complexity Comparison

| Method | Time Complexity | Memory |
|---|---|---|
| Isolation Forest | O(n log n) | Low |
| One-Class SVM | $O(n^2)$ | Medium |
| Baseline Authentication | O(1) | Very Low |

Although anomaly detection introduces overhead, the security gain significantly outweighs computational cost.

## XVI. Extended Experimental Discussion

Additional experiments explored sensitivity under varying noise conditions.

Noise levels from 1 percent to 10 percent were introduced.

Results showed moderate noise did not significantly degrade anomaly detection performance.

This suggests robust feature selection.

Detection performance remained above 90 percent under moderate perturbation.

This supports practical deployment viability.

TABLE VI: Detection Under Noise Variation

| Noise Level | Detection Rate | FPR |
|---|---|---|
| 1% | 95.1% | 3.6% |
| 3% | 94.3% | 4.1% |
| 5% | 93.6% | 4.2% |
| 10% | 91.7% | 5.3% |

### A. Feature Contribution Analysis

Different behavioral features contributed differently to attack detection.

Most informative:

- Response entropy
- Reliability variance
- Hamming distance patterns

Combining features produced stronger discrimination than any single feature alone.

## XVII. Application Scenarios

The proposed framework has broad relevance across IoT applications.

### A. Smart Healthcare

Medical IoT devices require strong lightweight authentication due to safety-critical operation.

Behavioral anomaly monitoring can strengthen device trust.

### B. Industrial IoT

Industrial cyber-physical systems require scalable authentication under resource constraints.

The proposed architecture is suitable for distributed sensor networks.

### C. Autonomous Systems

Autonomous platforms require resilient hardware-rooted identity.

Hybrid anomaly monitoring provides additional assurance.

### D. Edge Security Architectures

Because anomaly models remain lightweight, deployment at edge nodes becomes practical.

This aligns with emerging edge-native security architectures.

## XVIII. Limitations and Research Challenges

Despite encouraging results, limitations remain.

Current validation is simulation-based.

Hardware experiments remain future work.

Feature engineering may require adaptation for alternative PUF architectures.

Environmental effects in real silicon may introduce additional variability.

Future research should explore:

- FPGA validation
- Real silicon characterization
- Adaptive online learning
- Side-channel integrated anomaly detection

Addressing these challenges may further improve deployment readiness.

## XIX. Detailed Security Evaluation

To further analyze robustness of the proposed framework, a broader security evaluation was conducted across multiple attack classes and operating conditions.

### A. Attack Surface Analysis

The authentication system faces threats at several levels.

At the communication layer, attackers may attempt interception, replay, or challenge manipulation.

At the machine learning layer, adversaries may attempt model approximation or anomaly evasion.

At the device layer, environmental perturbation may impact reliability.

These threat surfaces motivate multi-layer protection.

Unlike single-point defenses, the proposed architecture introduces security redundancy through independent verification signals.

### B. Modeling Attack Resistance Analysis

Machine learning attacks attempt to approximate the PUF transfer behavior.

However successful response prediction alone does not guarantee successful impersonation.

The anomaly detector evaluates statistical properties of response behavior over authentication sessions.

This creates an additional attack burden.

To succeed, an adversary must reproduce:

- Correct predicted responses
- Genuine entropy characteristics
- Authentic reliability distributions
- Consistent temporal behavior

This significantly raises attack complexity.

### C. Replay Resistance Analysis

Replay attacks remain important in practical authentication systems.

The proposed framework improves replay resistance using:

- Session freshness through nonces
- Timestamp consistency
- Behavioral feature monitoring
- Statistical anomaly scoring

This combined approach reduces replay feasibility.

## XX. Sensitivity and Parameter Analysis

System sensitivity was evaluated under varying detection thresholds.

Threshold tuning impacts both false positives and detection rates.

Moderate thresholds produced the best tradeoff.

This demonstrates tunable operation depending on security requirements.

TABLE VII: Threshold Sensitivity Analysis

| Threshold | Detection | FPR |
|---|---|---|
| Strict | 96.2% | 6.1% |
| Moderate | 93.6% | 4.2% |
| Relaxed | 89.4% | 2.8% |

### A. Feature Sensitivity

Feature importance was analyzed experimentally.

Entropy and reliability variance contributed most strongly to anomaly separation.

Combining multiple features consistently outperformed single-feature detection.

## XXI. Comparative Discussion Against Conventional Authentication

Traditional lightweight authentication often relies solely on challenge-response verification.

Such approaches may fail when modeled responses appear valid.

The proposed architecture differs fundamentally by integrating behavioral intelligence.

Comparison advantages include:

- Higher resilience against learned attacks
- Adaptive anomaly awareness
- Lower dependence on stronger hardware redesign
- Lightweight computational feasibility

These properties make the approach attractive for practical deployment.

TABLE VIII: Security Comparison

| Scheme | Replay | Modeling | Adaptive |
|---|---|---|---|
| Conventional PUF | Weak | Weak | No |
| Controlled PUF | Medium | Medium | No |
| Proposed Method | Strong | Strong | Yes |

## XXII. Large Scale Deployment Considerations

Real-world IoT deployments may include thousands or millions of devices.

Authentication mechanisms must therefore scale.

The proposed framework supports scalable deployment through lightweight inference.

Potential deployment models include:

- Edge-hosted anomaly inference
- Distributed gateway validation
- Hybrid cloud-edge monitoring
- Federated security analytics

These models support scalable secure IoT ecosystems.

### A. Edge Resource Feasibility

Resource feasibility was examined.

Estimated requirements remain practical even for constrained devices.

Results support practical implementation.

TABLE IX: Estimated Resource Cost

| Metric | Proposed | Typical IoT Budget |
|---|---|---|
| Memory | 32KB | 64KB |
| Latency | 5ms | 20ms |
| CPU Overhead | Low | Moderate |

## XXIII. Broader Research Impact

This work contributes beyond PUF security alone.

It illustrates how lightweight artificial intelligence may augment hardware security primitives.

This concept may generalize to:

- Hardware trust anchors
- Device attestation
- Embedded anomaly monitoring
- Zero-trust edge security

This broader perspective increases relevance of the proposed approach.

## XXIV. Extended Future Research Directions

Several additional research opportunities remain.

### A. FPGA Prototype Validation

Future work should implement the framework on FPGA platforms for hardware validation.

### B. Adaptive Online Learning

Dynamic anomaly boundaries may improve long-term robustness.

### C. Side-Channel Integration

Behavioral monitoring may be combined with side-channel indicators for stronger detection.

### D. Federated Security Models

Federated anomaly learning may support collaborative IoT defense while preserving privacy.

### E. Deep Learning Resistant Extensions

Future research may examine stronger protection against advanced adversarial learning attacks.

These directions provide a path toward practical deployment and future publication-quality extensions.

## XXV. Summary Discussion

Across modeling attacks, replay attacks, scalability analysis, and sensitivity evaluation, results consistently support the proposed framework.

The major insight is that behavioral anomaly monitoring can strengthen hardware-rooted security without expensive hardware redesign.

This provides a practical path toward scalable lightweight IoT authentication.

## XXVI. Case Study and Practical Use Scenario

To further illustrate practical relevance, a representative IoT authentication scenario is considered.

Assume a distributed smart healthcare monitoring network containing wearable sensors, gateway nodes, and cloud-connected monitoring services.

Each sensor contains an embedded PUF used for lightweight device authentication.

During normal operation, devices produce characteristic challenge-response behavior and stable statistical authentication patterns.

Under a machine learning modeling attack, an adversary may generate forged responses approximating legitimate device outputs.

Traditional challenge-response verification alone may fail to detect sophisticated forgeries.

However the proposed behavioral anomaly detection layer monitors:

- Entropy consistency
- Reliability variation
- Temporal authentication behavior
- Statistical feature deviations

Even when forged responses appear valid, abnormal behavioral patterns can trigger anomaly detection.

This layered defense improves trust in medical device authentication where safety-critical operation is essential.

A similar approach applies to industrial IoT networks.

For example, distributed industrial sensors may use PUF authentication to protect control systems.

Compromise of a single node could threaten larger cyber-physical infrastructure.

Behavioral anomaly monitoring provides an additional security layer against impersonation and replay attempts.

### A. Practical Benefits

Practical advantages include:

- Lightweight deployment compatibility
- Improved authentication robustness
- Minimal hardware redesign requirements
- Scalability for large IoT ecosystems

These properties make the proposed approach attractive beyond simulation study alone.

### B. Broader Implications

The framework also suggests a broader security principle:

Combining hardware-rooted trust with lightweight intelligent monitoring may provide stronger security than either mechanism alone.

This concept may extend beyond PUF authentication to future embedded trust architectures.

This case study reinforces the practicality and broader applicability of the proposed framework.

TABLE X: Illustrative Deployment Benefits

| Criterion | Traditional PUF | Proposed Framework |
|---|---|---|
| Attack Detection | Moderate | High |
| Adaptability | Low | High |
| Scalability | Medium | High |
| Hardware Overhead | Low | Low |

## XXVII. Expanded Comparative Evaluation

To further evaluate the proposed framework, a broader comparative study was considered against representative authentication strategies.

Three approaches were compared:

- Conventional PUF-only authentication
- Hardened controlled PUF architectures
- Proposed anomaly-assisted authentication

Comparison criteria included security robustness, computational efficiency, adaptability, and deployment practicality.

TABLE XI: Comparative Evaluation

| Metric | Conventional | Controlled | Proposed |
|---|---|---|---|
| Modeling Resistance | Medium | High | High |
| Replay Resistance | Low | Medium | High |
| Adaptability | Low | Low | High |
| Computation Cost | Low | High | Low |
| Scalability | Medium | Medium | High |

Results suggest the proposed framework offers a favorable balance between security and lightweight implementation.

Unlike approaches requiring significant architectural redesign, the behavioral monitoring layer strengthens security without major hardware modification.

### A. Discussion of Tradeoffs

Every security design introduces tradeoffs.

Stronger PUF constructions may improve intrinsic resistance but often increase implementation complexity.

The proposed framework instead adds software-level intelligence while preserving lightweight hardware properties.

This makes it especially attractive for constrained IoT devices.

### B. Reliability Considerations

Reliability remains important for practical authentication.

Environmental variation may affect raw PUF responses.

However behavioral feature analysis can remain robust even under moderate response variability.

This improves resilience under real-world operating conditions.

Comparative Performance Radar Chart Placeholder

Fig. 3: Illustrative comparison of security-performance tradeoffs.

The comparative evaluation further supports the practicality of the proposed architecture.

## XXVIII. Broader Conclusion Perspective

Beyond improving PUF authentication specifically, this work demonstrates a broader research direction:

Lightweight artificial intelligence can augment hardware trust mechanisms.

Rather than viewing AI and hardware security as separate domains, their integration may enable stronger embedded security architectures.

This perspective has implications for future secure edge systems, zero-trust device ecosystems, and resilient IoT infrastructures.

## XXIX. Conclusion

This paper presented a lightweight AI-assisted behavioral anomaly detection framework for securing PUF-based IoT authentication.

Results showed:

- Greater than 90 percent attack detection
- Low false positives
- Lightweight overhead
- Improved attack resilience

The approach provides scalable protection for next-generation IoT authentication.

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